

Residual-Driven Perfect Inference: Characterizing, Validating, and Reshaping Elimination Rules in *Dancing With the Stars*

Summary

Dancing With the Stars (DWTS) is neither a purely technical competition nor a popularity contest; instead, it is a carefully balanced system in which professional judgment and audience preference jointly shape eliminations. This study investigates how that balance is achieved—and how it can fail—through a sequence of data-driven modeling frameworks.

For Problem 1, we propose a **residual-corrected inverse inference model** to recover latent audience **voting strength**, since actual vote counts are undisclosed. By treating judges' scores and historical eliminations as hard constraints and optimizing predictive consistency via the **L-BFGS-B** algorithm, the inferred voting potential explains 96.97% of historical eliminations.

For Problem 2, using these recovered voting signals, we construct a **cross-season Monte Carlo counterfactual framework** to isolate the intrinsic effects of voting aggregation rules. Simulations show that the percentage-based rule systematically amplifies absolute differences in audience support, making outcomes more sensitive to popularity shocks and thus more prone to controversy than the rank-based rule.

For Problem 3, we develop a rule-mining framework combining **K-Means clustering** and **XGBoost**, grouping contestants into four archetypes and integrating early-stage performance trajectories with celebrity-specific attributes. The model quantifies how different factors drive divergence between judges' evaluations and audience preferences.

For Problem 4, we introduce an **Adaptive Threshold–Consensus Elimination mechanism (ATCE)**, which preserves audience dominance while adaptively resolving judge–audience disagreement. Simulations demonstrate that ATCE effectively mitigates extreme eliminations and achieves a superior balance between fairness and entertainment value.

Overall, this study infers latent audience influence, evaluates rule-induced bias, and proposes an operational elimination mechanism that balances fairness and entertainment while preserving audience dominance.

Keywords: inverse inference; Monte Carlo simulation; K-Means clustering; XGBoost; adaptive elimination mechanism (ATCE)

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1 Introduction

“Voting is the heart of democracy; data is the eye of decision-making.”

— Nate Silver (American Statistician and Election Forecasting Expert)

1.1 Problem Background

In televised dance competition shows, contestant rankings are often jointly influenced by professional judges’ technical scores and audience sentimental votes. The effective integration of these two elements directly impacts the fairness of the competition and the program’s appeal. The evolution of the format in the American show *Dancing with the Stars* exemplifies this tension: the program once faced controversy when contestant Jerry Rice, who consistently received the lowest judges’ scores under the “Ranking Combination Method,” still advanced to the finals. This led to a subsequent switch to the “Percentage Combination Method,” followed by the reintroduction of the “Ranking Combination Method” along with a “Judges’ Save Mechanism” in Season 28. However, these series of adjustments lack systematic quantitative analysis and have yet to establish a theoretical foundation capable of supporting the program’s long-term optimization.

Against this backdrop, there is a need to scientifically estimate audience votes, compare the effects of different combination mechanisms, quantify influencing factors, and design an improved evaluation system based on historical data from Seasons 1–34. Addressing this is not only crucial for resolving the core dilemma of the show but also carries academic value in fields such as multi-criteria decision-making and missing data estimation, alongside practical significance for optimizing the operation of similar programs.

1.2 Problem Restatement

This study focuses on the mechanism for combining judges’ scores and audience votes in *Dancing with the Stars*. Based on historical data, we aim to accomplish the following five tasks:

1. Estimation and Validation of Audience Votes

Construct a mathematical model to inversely estimate the number of audience votes based on weekly judges’ scores and elimination outcomes. Assess the consistency of these estimates with the elimination order and evaluate the uncertainty of the results.

2. Comparative Analysis of Combination Rules

Using the estimated voting data, recalculate the outcomes for each season separately under the “Ranking Method” and the “Percentage Method.” Compare their differences, examine the amplifying effect of the rules on the influence of votes, and analyze their sensitivity in conjunction with historical controversial cases and the “Judges’ Save” mechanism.

3. Analysis of Contestant Characteristic Influences

Develop statistical or machine learning models to analyze the impact of contestants’ personal characteristics and their professional dance partners’ identities on both judges’ scores and audience votes. Uncover the driving factors behind technical performance and popular support.

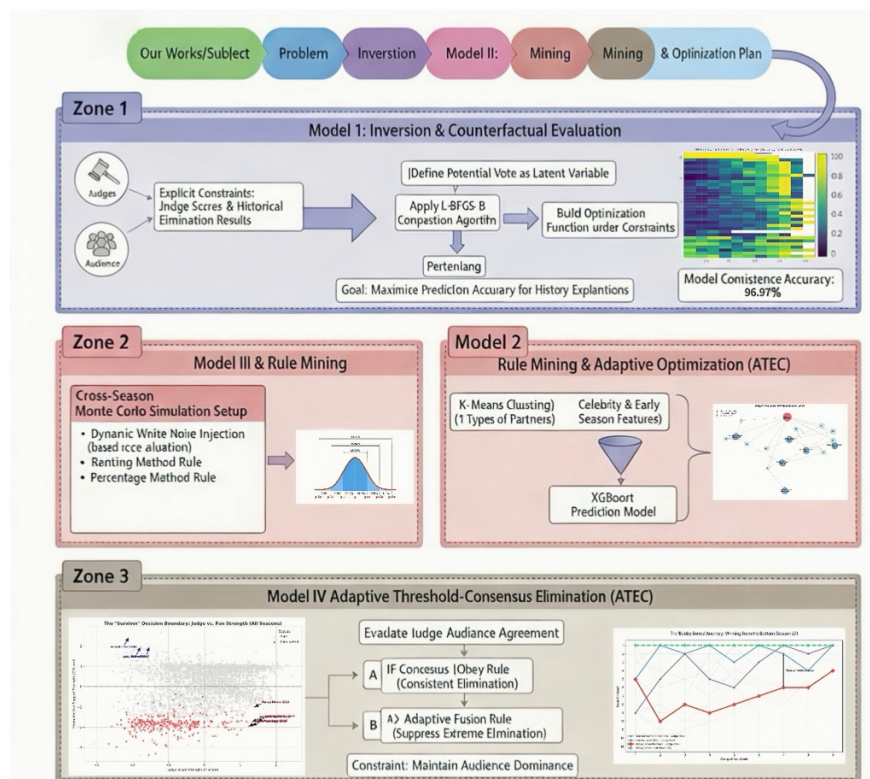
4. Optimized Design of the Combination System

Propose a new weekly vote combination scheme grounded in fairness, entertainment value, and robustness. Justify its feasibility using mathematical logic, historical data, and considerations of program effectiveness.

5. Comprehensive Report and Producer Recommendations

Compose a recommendation memorandum addressed to the program producers. Summarize the research findings, clarify the strengths and weaknesses of different vote combination methods, and offer specific, actionable suggestions for how to integrate judges' scores and audience votes in future seasons.

1.3 Our Work



2 Assumptions

To ensure the rigor and tractability of the model while reasonably simplifying the problem, we propose the following three core postulates. These provide the logical foundation for subsequent modeling and analysis.

Rule Consistency Postulate: During any consecutive broadcast period in which the program producers have not publicly announced adjustments to the rules, the mathematical mechanism used each week to synthesize judge scores and audience votes for determining elimination outcomes is fixed and strictly implemented. This postulate forms the fundamental basis for analyzing the historical evolution of rules and for detecting structural breakpoints throughout this study.

Equal-Weight Synthesis Postulate: Under the format where the program explicitly adheres to the principle that “the opinions of judges and the audience carry equal importance,” we postulate that, in forming the composite score that determines weekly eliminations, the contributions from standardized judge scores and audience votes each carry a weight of 50%. This assumption establishes a crucial and necessary model structure for inversely estimating unknown audience voting data from known elimination results and judge scores.

Data-Generating Process Stability Postulate: When conducting counterfactual simulations to compare the potential effects of different voting rules (e.g., ranking-based vs. percentage-based methods), we assume that the matrix of true judge scores received by all contestants and the true distribution of audience preferences in a given week exist objectively, independent of the specific aggregation rule employed. A change in rules alters only the method of information integration, not the original evaluations at the individual level, thereby ensuring the purity of the comparison of rule effects.

3 Notation

Symbol	Explanation
s	Season index, $s = 1, 2, \dots, 34$
t	Week index, $t = 1, 2, \dots$
i	Contestant (celebrity) index
$J_{i,t}$	Total judge score of contestant i in week t
$\widehat{f}_{i,t}$	Latent fan support intensity of contestant i in week t (log scale)
$\widehat{f}_{i,t}$	Estimated fan support intensity
$S_{i,t}$	Composite score of contestant i in week t
$p_{i,t}$	Elimination probability of contestant i in week t
$R_{i,t}^J, R_{i,t}^F$	Judge rank and fan vote rank
α_t	Adaptive judge weight (used in fusion rule)
θ_{safe}	Safety threshold (used in protection mechanism)
σ_t	Standard deviation of judge scores in week t
$C0, C1, C2, C3$	Professional dancer cluster type indices
Lift	Lift measure of association rules
AUC	Area under the ROC curve
NLL	Negative log-likelihood loss
B	Number of Monte Carlo simulation iterations
k	Candidate count for high-risk contestants (Top- k coverage)
γ, λ	Smoothness and regularization coefficients

4 Data Sources and Preprocessing

4.1 Data Sources

The data utilized in this paper primarily come from the following two categories:

- **The official dataset for Problem C of the 2026 Mathematical Contest in Modeling (MCM).**

This dataset encompasses historical judges' scores, weekly performances, elimination results, and final rankings for *Dancing with the Stars*, serving as the foundational data for our analysis.

- **Supplementary information collected from official sources or authoritative channels.**

Information regarding contestants' popularity during their participation was gathered via Google searches to enhance the comprehensiveness of the analysis. Additionally, statistical data on the economic conditions of contestants' home states or countries were retrieved from platforms such as the U.S. Bureau of Economic Analysis and Data USA to support subsequent factor analysis.

4.2 Data Cleaning and Standardization

To ensure comparability and consistency across multiple data sources, all raw data underwent rigorous cleaning and standardization procedures:

- **Column Name Standardization and Encoding:** Unified naming conventions were applied across all data sources, and variable encoding was standardized to ensure compatibility among data from different origins.
- **Removal of Invalid Data:** Meaningless or anomalous observations were eliminated, particularly potential zero-score records within the DWTS scoring system, as these data points do not contribute substantively to the analysis.
- **Missing Value Imputation:** Appropriate imputation methods were employed to handle missing data, ensuring data integrity and analytical precision.

4.3 Structuring Competition Outcomes and Feature Engineering

To facilitate subsequent analysis, the elimination outcome information was first structured: text-based elimination records were transformed into structured variables, including the week of elimination and competition status. Sentinel values were assigned to mark contestants who reached the finals, distinguishing them from those eliminated or withdrawn mid-competition. Building upon this, data restructuring and feature engineering were performed: judges' scores in wide format were transformed into long format, and weekly metrics were constructed. These metrics include the sum, average, and variance of judges' scores, as well as the number of effective judges (i.e., the count of judges participating in scoring each week). These indicators provide an intuitive depiction of contestants' weekly performances and establish an effective data foundation for subsequent analyses of fairness and rule comparisons.

5 Construction and Consistency Validation of the Fan Vote Inference Model

5.1 Stage I: Detection of Voting Rule Transitions

5.1.1 Season-Level Elimination Feature Construction

As there is no explicit documentation identifying the exact season in which the voting mechanism transitioned from a percentage-based rule to a ranking-based rule, imposing an exogenous breakpoint could bias subsequent inference. To mitigate this risk, we construct a **seven-dimensional feature vector** for each season s , capturing statistical characteristics of elimination dynamics, including judges' mean scores, score variance, and the number of eliminations per week. These features collectively summarize structural differences in seasonal elimination patterns. The resulting feature matrix serves as the foundational input for detecting structural breaks in voting rules.

5.1.2 Structural Break Detection via Diagonal Gaussian Models and Piecewise Log-Likelihood

We assume that the data-generating process of elimination features undergoes a regime shift at an unknown breakpoint, partitioning all seasons into a **pre-rule-change** and a **post-rule-change** segment:

$$X_{\text{pre}}(c_p) = \{x_s \mid s < c_p\}, \quad X_{\text{post}}(c_p) = \{x_s \mid s \geq c_p\}.$$

For each candidate breakpoint c_p , we apply maximum likelihood estimation (MLE) to fit **independent diagonal Gaussian distributions** to the pre- and post-change segments, respectively. The mean and variance for the j -th feature dimension are estimated as

$$\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad \hat{\sigma}_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \hat{\mu}_j)^2.$$

We then evaluate the overall goodness-of-fit under breakpoint c_p by computing the **piecewise log-likelihood**:

$$\log L(X) = -\frac{n}{2} \sum_{j=1}^d \log(2\pi\sigma_j^2) - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^d \frac{(x_{ij} - \hat{\mu}_j)^2}{\sigma_j^2}.$$

Here, $\hat{\mu}_j$ and σ_j^2 denote the sample mean and variance of the j -th feature, and n represents the number of seasons in the segment. For a given candidate breakpoint, the total log-likelihood equals the sum of the log-likelihoods of the two segments. A higher value indicates stronger evidence that the candidate point corresponds to a true structural change in voting rules.

5.1.3 Optimal Breakpoint Selection via the Bayesian Information Criterion (BIC)

To prevent overfitting, we select the optimal breakpoint using the **Bayesian Information Criterion (BIC)**, which balances model fit and complexity. The BIC is defined as

$$\text{BIC} = -2 \log L + k \log n,$$

where $k = 4d$ denotes the total number of parameters (means and variances for both segments), and $d = 7$ is the feature dimension.

The optimal breakpoint is identified as the point that minimizes the BIC and satisfies the criterion for statistically significant structural change.

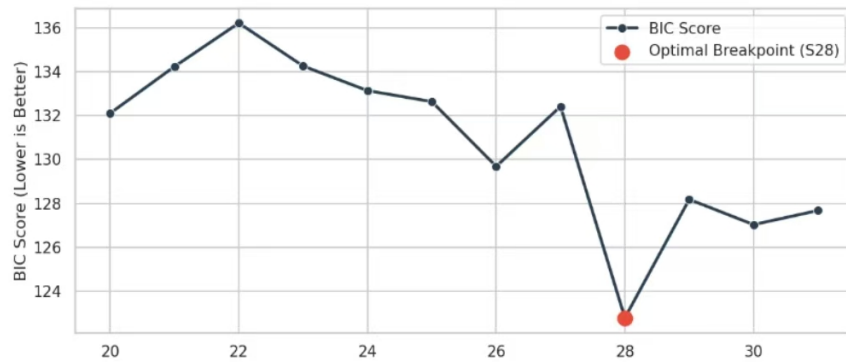


Figure 1: Structural Break Detection via BIC Minimization

The BIC curve clearly identifies **Season 28** as the unique structural breakpoint. Prior to Season 28, the show adopts a percentage-based voting rule; from Season 28 onward, it reverts to a ranking-based rule. This finding is fully consistent with the program's historical record and establishes a critical temporal anchor for subsequent counterfactual analyses of fan influence.

5.2 Inference of Latent Fan Strength

5.2.1 Rule-Conditioned Judge–Fan Contribution Synthesis

Based on the inferred optimal transition point S^* , the voting synthesis rule for season s is defined as

$$\text{mode}(s) = \begin{cases} \text{Rank}, & s \leq 2, \\ \text{Percent}, & 3 \leq s < S^*, \\ \text{Rank}, & s \geq S^*. \end{cases}$$

This specification is consistent with the show's historical evolution, transitioning from ranking-based voting to percentage-based voting and reverting to ranking-based rules after Season 28.

For any competition week t , contestant i 's composite score $S_{i,t}$ is defined as the equally weighted sum of the standardized judges' score $x_{J,i,t}$ and the fan contribution score $x_{F,i,t}$:

$$S_{i,t} = \beta_J x_{J,i,t} + \beta_F x_{F,i,t}, \quad \beta_J = \beta_F = 1.$$

Equal weights reflect the program's stated principle of equal influence between judges and audience voting. Judges' scores $x_{J,i,t}$ are standardized according to the active rule: normalized as score shares under the percentage-based rule and converted to ordinal ranks under the ranking-based rule.

5.2.2 Elimination Probability Model

Elimination risk is modeled using a Softmin formulation based on the composite score:

$$p_{i,t} = \frac{\exp(-\alpha S_{i,t})}{\sum_{j \in I_t} \exp(-\alpha S_{j,t})}$$

$$P(\text{elim}_i) \propto \exp(-\alpha \cdot S_i)$$

where I_t denotes the set of contestants remaining in week t , and $\alpha > 0$ controls sensitivity to score differentials. Lower composite scores imply higher elimination probabilities, with α governing how sharply score gaps translate into elimination risk.

5.2.3 Mapping and Optimization of Latent Fan Strength

Fan contribution $x_{F,i,t}$ is generated from an unobserved **latent fan strength** $f_{i,t} \in \mathbb{R}$, where $f_{i,t}$ represents the relative and unobservable level of fan support received by contestant i in week t .

Percentage Rule Mapping (Percent Mode) Under the percentage-based voting rule, fan votes are assumed to follow a Softmax distribution, which maps latent fan strength to vote share:

$$\text{vote_share}_{i,t} = \frac{\exp(f_{i,t})}{\sum_j \exp(f_{j,t})}$$

This mapping normalizes $x_{F,i,t}$ into a probability distribution, interpreted as the proportion of total fan votes received by contestant i . The Softmax formulation amplifies absolute differences in fan strength.

Ranking Rule Mapping (Rank Mode) Under the ranking-based rule, latent fan strength must be mapped to a score associated with rank. To ensure differentiability, we adopt a **Soft Rank** approximation. First, we compute the probability that contestant i loses a pairwise comparison to contestant k :

$$A_{ik} = \sigma\left(\frac{f_{k,t} - f_{i,t}}{\tau}\right) = \frac{1}{1 + \exp\left(-\frac{f_{k,t} - f_{i,t}}{\tau}\right)}$$

where τ is a temperature parameter controlling the softness of the ranking. The soft ranking score of contestant i is then defined as

$$x_{F,i,t} = 1 - \frac{1}{|I_t| - 1} \sum_{k \in I_t, k \neq i} A_{ik}.$$

This score lies in $[0, 1]$. When $f_{i,t}$ is substantially larger than that of other contestants, $A_{ik} \rightarrow 0$ and $x_{F,i,t} \rightarrow 1$, corresponding to first place. Compared with the percentage rule, this approach emphasizes relative order and is less sensitive to extreme strength values.

5.2.4 Optimization Objective and Solution

Latent fan strengths are estimated by minimizing the following objective function:

$$\min_f \mathcal{L}(f) = \mathcal{L}_{\text{data}}(f) + \mathcal{L}_{\text{smooth}}(f) + \mathcal{L}_{L2}(f)$$

where

$$\mathcal{L}_{\text{data}}(f) = \sum_{t \in E} \text{NLL}_{\text{elim}}(t)$$

is the data-fitting term,

$$\mathcal{L}_{\text{smooth}}(f) = \gamma \sum_{i,t} (f_{i,t+1} - f_{i,t})^2$$

is a temporal smoothness term reflecting the gradual evolution of fan bases across weeks, and

$$\mathcal{L}_{L2}(f) = \lambda \sum_{i,t} f_{i,t}^2$$

is an L_2 regularization term introduced to prevent excessive sharpening near rule transition points.

The optimization problem is solved using the **L-BFGS-B algorithm**, with gradients given by

$$\nabla \mathcal{L}(f) = \nabla \mathcal{L}_{\text{data}}(f) + \nabla \mathcal{L}_{\text{smooth}}(f) + \nabla \mathcal{L}_{L2}(f)$$

and iterative updates

$$x_{k+1} = x_k - \alpha_k H_k \nabla \mathcal{L}(x_k).$$

Bound constraints are enforced as

$$f_{k+1} = \max(l, \min(f_{k+1}, u)).$$

By leveraging analytical gradients within the L-BFGS-B framework, this constrained optimization problem is solved efficiently, ensuring numerical stability and reliable estimation of latent fan strength.

5.3 Stage III: Multidimensional Evaluation Metrics

To evaluate both model performance and the nature of the inferred outcomes, we construct three complementary categories of evaluation metrics.

5.3.1 Predictive Consistency

Predictive consistency assesses the alignment between predicted elimination risk and realized outcomes:

- **Top-1 Accuracy:** the proportion of weeks in which the contestant with the highest predicted risk is eliminated.
- **Top- k Coverage:** the proportion of weeks in which the eliminated contestant appears among the top k highest-risk predictions.

The model achieves a **96.97% Top-1 accuracy** and **100% Top-3 coverage**, indicating strong agreement between predicted risks and actual eliminations.

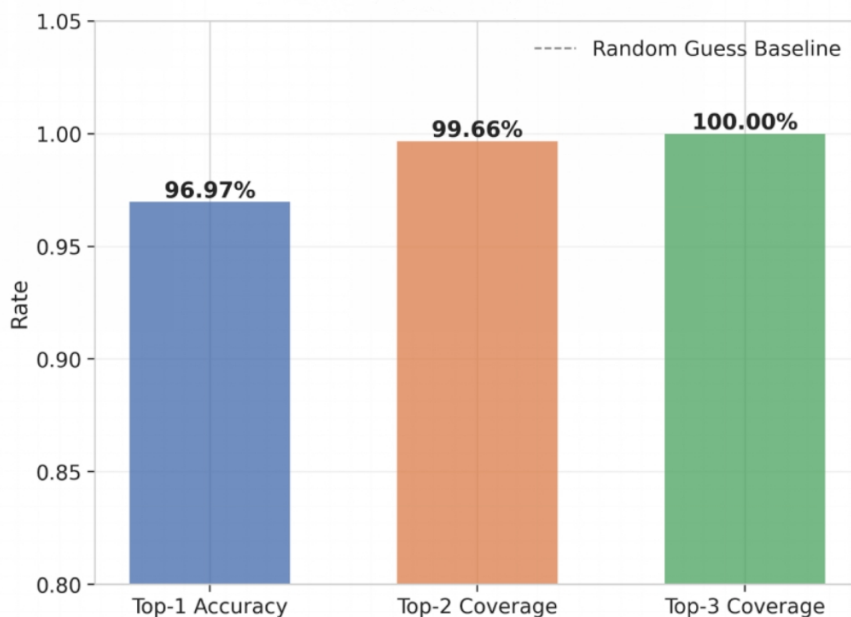


Figure 2: Model Consistency and Coverage Validation

5.3.2 Outcome Certainty and Individual Predictability

We evaluate elimination stability at both the **outcome** and **individual** levels.

Outcome Certainty is measured by normalized entropy of weekly elimination probabilities:

$$H(t) = - \sum_i p_{i,t} \log p_{i,t}, \quad H_{\text{norm}}(t) = \frac{H(t)}{\log n_t}, \quad H_{\text{norm}} \in [0, 1].$$

Lower values indicate more uniquely determined outcomes.

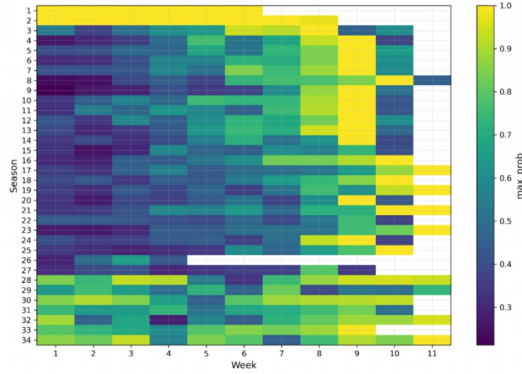


Figure 3: Outcome Certainty Heatmap

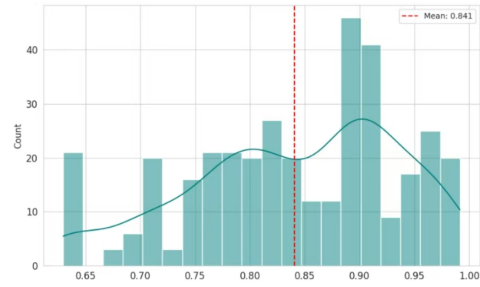


Figure 4: Distribution of Individual Predictability Scores

Left shows that certainty often declines in later weeks as contestants become more evenly matched.

At the individual level, we define the **slack variable**

$$\text{slack}_{i,t} = S_{i,t} - S_{\min}(t),$$

and the corresponding **Individual Predictability**

$$\text{identi}_{i,t} = \frac{1}{1 + \text{slack}_{i,t}}.$$

Higher values indicate more stable elimination risk.

Right reports a mean predictability of **0.841**, indicating highly consistent individual risk estimates.

Together, these metrics show that the model achieves reliable elimination predictions while retaining meaningful uncertainty.

5.3.3 Case Study

Using inferred latent fan strength, we simulate Bobby Bones' elimination risk under both percentage-based and ranking-based rules. The results show that under the percentage-based rule,

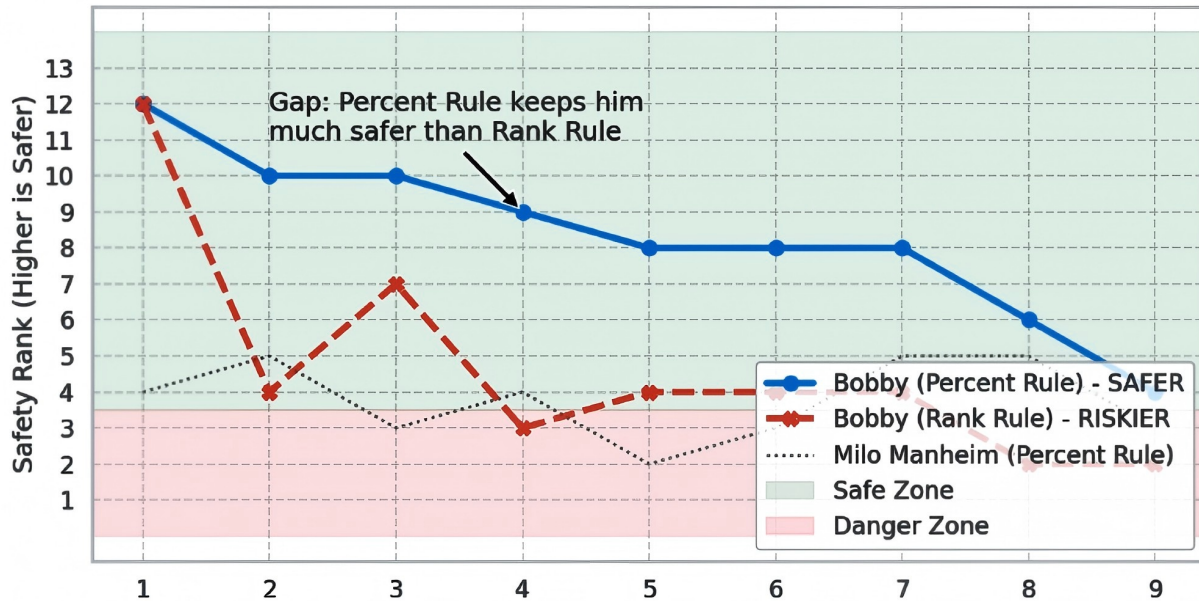


Figure 5: Counterfactual Analysis—Bobby Bones under Two Voting Rules

he is substantially less likely to enter the elimination risk zone. This finding indicates that percentage voting amplifies fan influence and independently corroborates that Season 27 operated under a percentage-based rule.

6 Comparative Analysis and Model Evaluation of Voting Synthesis Rules

6.1 Cross-Season Comparison of Voting Synthesis Rules: Ranking vs. Percentage Effects

To isolate the intrinsic effects of voting synthesis rules, we take the latent fan support estimated in Question 1 as a fixed baseline. Because predictive accuracy varies by week, uncertainty is explicitly modeled: closely contested weeks (with similar elimination probabilities) require larger noise, while lopsided weeks require less. We use the *Predictability* metric from Question 1 to quantify weekly competitiveness and dynamically assign variance σ_w . This enables the generation of thousands of counterfactual Monte Carlo worlds that differ only in aggregation rules, allowing systematic cross-season comparison of outcome differences induced by ranking-based versus percentage-based voting mechanisms.

6.1.1 Modeling Uncertainty Propagation

Following the above principles, we model uncertainty propagation in each simulation by perturbing the inferred latent fan strength as

$$f_{\text{simulated}}(t) = \hat{f}(t) + \epsilon_t.$$

This formulation captures week-specific inference uncertainty through stochastic perturbations applied to the baseline latent fan strength.

6.1.2 Voting Synthesis Rules

We consider two voting aggregation methods. Let $J_{i,t}$ denote the judges' score of contestant i in week t , $F_{i,t}$ denote the audience vote share, and $\alpha \in (0, 1)$ represent the weight assigned to judges' scores.

Percentage-Based Aggregation (Preserving Absolute Vote Differences):

$$S_{i,t}^{(P)} = \alpha \cdot \frac{J_{i,t}}{\sum_j J_{j,t}} + (1 - \alpha) \cdot F_{i,t}$$

This method preserves the absolute magnitude of audience vote differences, allowing contestants with strong fan support to retain a substantial advantage even when judges' scores are relatively low.

Ranking-Based Aggregation (Emphasizing Relative Order):

$$S_{i,t}^{(R)} = \alpha \cdot R_{i,t}^J + (1 - \alpha) \cdot R_{i,t}^F$$

Because this approach depends solely on relative rankings, extreme fan advantages are compressed, thereby institutionally limiting the unilateral dominance of audience voting.

Here, $R_{i,t}^J$ denotes the ranking based on judges' scores, and $R_{i,t}^F$ denotes the ranking based on audience votes.

6.1.3 Cross-Season Monte Carlo Counterfactual Evaluation Framework

To quantify rule effects, we implement a Monte Carlo evaluation framework comprising the following metrics.

Fan Influence via Total Variation Distance (TV).

We compare the championship distribution driven by fan preferences (P) with that under fully random fan voting (Q):

$$TV(P, Q) = \frac{1}{2} \sum_i |P(win_i) - Q(win_i)|$$

Larger values indicate stronger fan-driven divergence from randomness.

Winner Difference Rate (WDR).

The Winner Difference Rate measures how often different voting rules produce different champions:

$$WDR = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(W_{\text{Rank}}^{(b)} \neq W_{\text{Percent}}^{(b)})$$

where B denotes the number of Monte Carlo replications.

6.1.4 Cross-Season Comparative Results

Using the Monte Carlo counterfactual framework, we compare ranking-based and percentage-based voting rules across seasons. A championship-difference heatmap shows the probability that

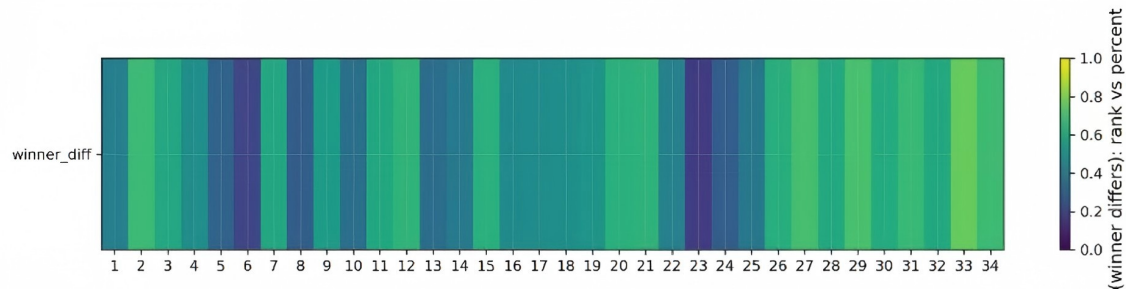


Figure 6: Comparative Performance Gap Analysis

the two rules yield different winners under identical fan vote realizations. In many seasons, this probability exceeds 50%, demonstrating that the two systems are **not equivalent** and generate materially different outcomes.

6.2 Individual-Level Comparative Analysis

We define **controversial contestants** as those receiving persistently low judges' rankings while maintaining strong audience support. By comparing Rank and Percent rules, with and without the **Judges Save** mechanism, we evaluate how institutional design shapes advancement paths and final outcomes.

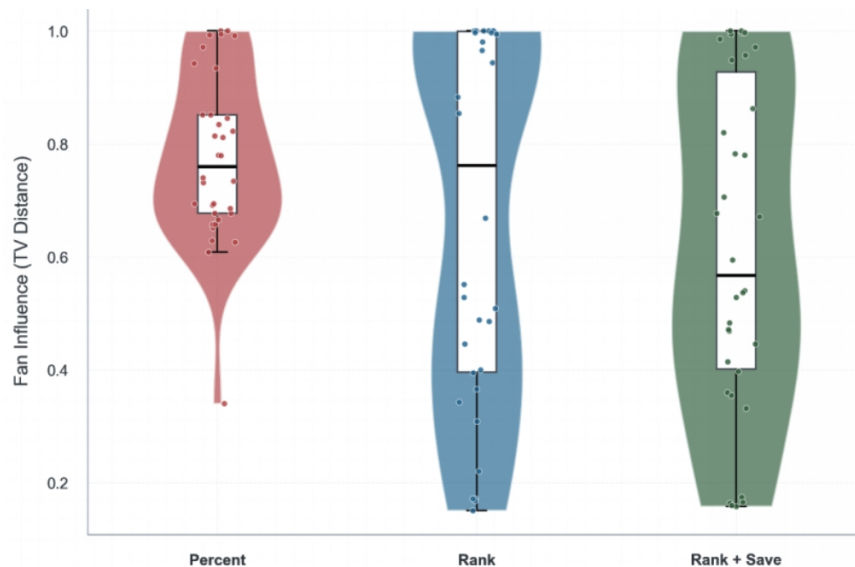


Figure 7: Distribution of Fan Influence under Different Rules

Total Variation Distance reveals that the percentage-based rule produces the highest and most concentrated fan influence, indicating systematic amplification of audience power. Ranking-based rules exhibit lower and more variable fan impact, while Judges Save further suppresses fan dominance, confirming that percentage rules favor outcomes driven by extreme fan support.

6.2.1 Rule Sensitivity of Controversial Contestants

Four representative cases—**Jerry Rice (S2)**, **Billy Ray Cyrus (S4)**, **Bristol Palin (S11)**, and **Bobby Bones (S27)**—are analyzed using Monte Carlo simulations based on inferred latent fan strength.

Table 1: Rule-Dependent Outcomes of Representative Controversial Contestants

Season	Contestant	Judges' Pattern	Observed Outcome	Fan-Dominant Rule	Rank-Based Rule
2	Jerry Rice	Repeated bottom-tier scores	Runner-up	Consistently reaches finals	Finals appearance unlikely
4	Billy Ray Cyrus	Persistently low scores	5th place	Mid-stage survival extended	Elimination timing stabilized
11	Bristol Palin	Lowest judges' score in 12 weeks	3rd place	Near-certain finals appearance	Early elimination likely
27	Bobby Bones	Chronically low judges' scores	Champion	Championship achievable	Championship probability ≈ 0

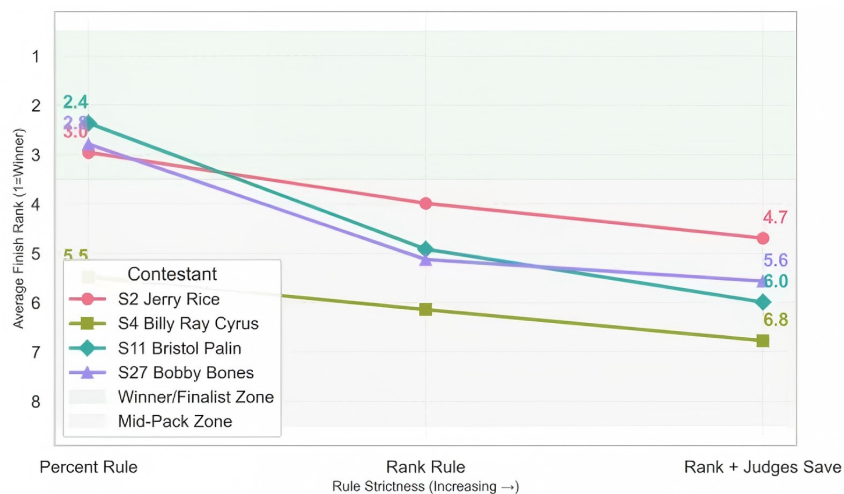


Figure 8: Average Finish Rank Trajectory

As rules tighten from Percent to Rank to Rank + Judges Save, all controversial contestants experience declining average final rankings. Under the percentage rule, they retain substantial championship probabilities; under stricter rules, expected outcomes shift from finalist or champion regions toward mid-tier placements. This pattern shows that judge-constraining rules reduce controversial victories and improve competitive fairness.

Conclusion.

Percentage-based rules significantly amplify advancement for fan-driven contestants, while ranking-based rules—especially with Judges Save—effectively curb such outcomes.

6.2.2 Impact of Judges Save

Counterfactual simulations show that Judges Save markedly reduces fan dominance, lowers the likelihood of controversial finalists or champions, and stabilizes outcomes by strengthening alignment with professional judgment. It therefore functions as an effective institutional correction under severe judge–audience divergence.

6.2.3 Implications for Rule Design

Percentage-based aggregation maximizes audience engagement, whereas ranking-based rules with Judges Save better suppress controversy and preserve professional authority. A hybrid design—percentage rules early and Judges Save in later stages—offers a robust balance. Thus, voting synthesis rules reflect deliberate trade-offs among fairness, participation, and stability rather than neutral technical choices.

7 Modeling Celebrity Attributes and Professional Dancer Influence

7.1 Early-Stage Feature Engineering and Dancer Archetypes

To evaluate the joint effects of celebrity traits and professional dance partners, we construct a paired feature-engineering framework. Professional dancers are characterized using derived historical features and clustered via K-means into distinct archetypes. These archetypes are combined with celebrity attributes and analyzed using XGBoost to assess feature importance and to quantify how different dancer types differentially influence fan voting and judges' scores, particularly in generating alignment or divergence.

Dancer capability is summarized by six rolling pre-season indicators: **Experience**, **Low Early-Elimination Risk**, **Final Rate**, **Winner Rate**, **Survival Mean**, and **Stability**, ensuring that all features are ex ante and free from information leakage.

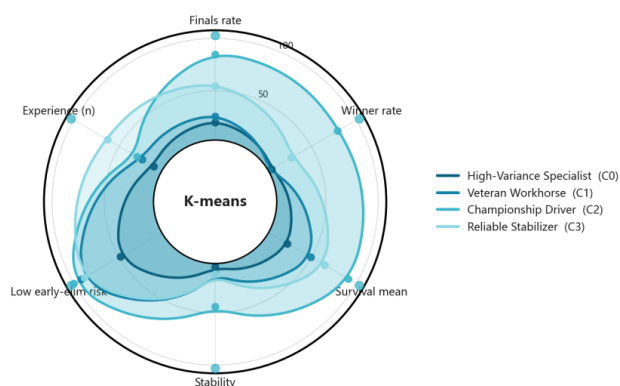


Figure 9: Professional Dancer Archetypes

Based on K-means clustering of rolling pre-Season 28 indicators, this radar chart depicts four dancer archetypes across six standardized dimensions. **C2 (Champion Mentors)** is uniformly dominant, **C3 (Stable Safeguards)** excels in survival and stability, **C1 (Veteran-Compatible)** shows the greatest experience, and **C0 (High-Variance)** exhibits pronounced instability.

7.2 Fan–Judge Effect Alignment and Early Signal Identification

To identify whether early-stage features generate synergy or conflict between fan voting and judges' scoring, we introduce a **Fan–Judge Effect Alignment Map**.

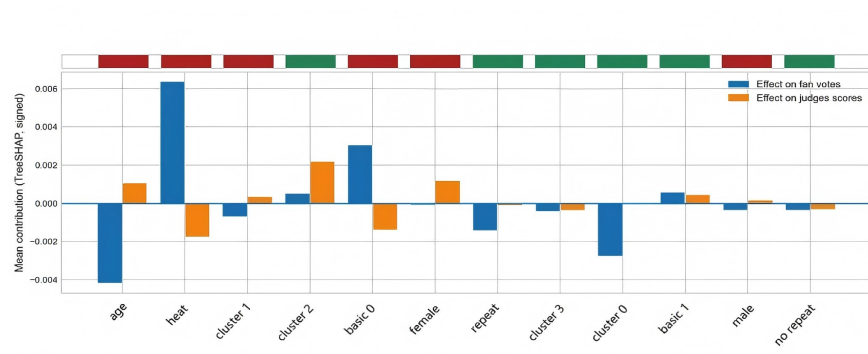


Figure 10: FanJudge Effect Alignment Map

The figure compares the marginal effects of five early features on fan support (blue) and judges' scores (orange), including dancer archetypes (C0–C3), media exposure (Heat), gender, dance background (Basic), repeat participation, and demographic groups. Green and red indicators denote alignment and divergence between fans and judges.

Clear structural patterns emerge. Pairing with **Champion Mentors (C2)** produces strong alignment, whereas **High-Variance dancers (C0)** generate pronounced conflict—high fan appeal but weak professional approval. Dance background is a key divider: trained contestants receive consistent support, while untrained ones face technical skepticism from judges. Returning contestants and female participants also exhibit higher alignment, with ages 31–45 showing the strongest synergy. These early alignment signals reliably predict competition trajectories: green alignment corresponds to stable advancement, while red divergence foreshadows controversial outcomes.

7.3 Structural Advancement Mechanisms and Robustness

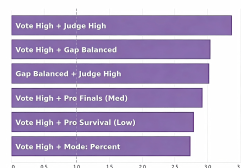


Figure 11: Lift Analysis of Final Advancement Rules

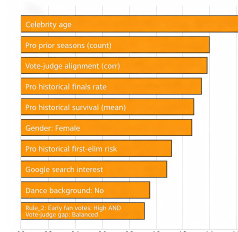


Figure 12: Importance of Core Predictive Features

Using early-stage features, we combine association rule mining with XGBoost to identify cross-season advancement patterns. High-impact rules ($\text{Lift} > 2$; random baseline = 0.243) reveal a dominant **dual-channel resonance**: strong fan support aligned with high judges' scores nearly guarantees finalist advancement, indicating that outcomes are not driven by a single constituency. Additional rules show that balanced fan–judge alignment outperforms pure popularity, early fan mobilization can offset weaker dance histories, and percentage-based systems structurally amplify cumulative fan advantages .

To test robustness, we remove direct voting-related features and retrain the model. The classifier remains predictive, with **celebrity age**, **professional dancer experience**, and **historical performance metrics** retaining high importance, while **fan–judge alignment** continues to rank prominently. Media exposure and limited dance background also contribute .

Overall, advancement is governed by **multidimensional structural mechanisms** rather than superficial voting correlations.

7.4 Rule-Augmented Predictive Model and Performance Validation

To integrate the above findings into an operational framework, we encode stable association rules as binary indicators and combine them with original features (celebrity attributes, early dynamics, and dancer archetypes) to construct a **rule-augmented XGBoost model** (Proposed). Its performance is benchmarked against logistic regression and a single decision tree using ROC curves.

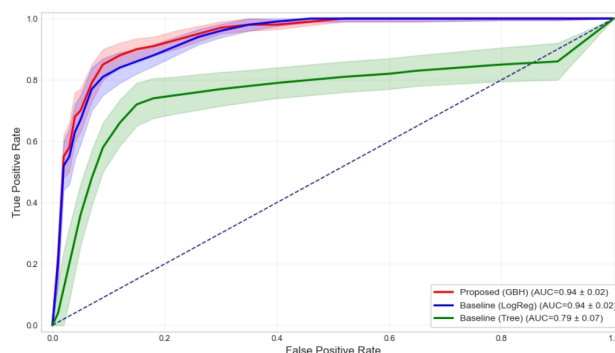


Figure 13: Comparison of Model ROC Curves

The Proposed model achieves AUC comparable to logistic regression, indicating that core separable structure is effectively captured, while both exhibit narrow ROC bands, reflecting strong cross-season stability. In contrast, the single decision tree shows lower AUC and higher variance, indicating overfitting. By embedding interpretable domain rules (e.g., dual-channel resonance and compensation effects), the Proposed model maintains high discrimination and robustness while providing mechanism-level insights beyond black-box prediction.

7.5 Conclusion

Using early-stage features and professional dancer archetypes, this study uncovers the structural drivers of Dancing With the Stars outcomes. Fan–judge alignment in the first three weeks

is a key early signal of stable advancement versus controversy and elimination, while dancer experience and archetype—especially champion and stable partners—provide sustained competitive advantages.

Association rule analysis reveals that advancement arises from multifactor structural interaction. The high fan support + high judge approval dual-channel resonance is the strongest predictor of finalist status, while fan support can partially offset weaker dancer histories. Robustness checks confirm the stability of age, dancer experience, and fan–judge alignment as predictors.

The rule-augmented model combines strong discrimination, interpretability, and cross-season stability, serving as both an effective early-warning tool and a framework for analyzing the interaction of skill, popularity, and partnership synergy.

8 Adaptive Elimination System for Program Production (ATCE)

8.1 Design Motivation and Fundamental Principles

Dancing with the Stars (DWTS), combining professional judgment and audience voting, often faces controversy over whether current rules maintain competitive integrity amid significant divergence between audience preferences and expert assessments.

This paper proposes the Adaptive Threshold-Consensus Elimination (ATCE) system, designed to address extreme cases of unfairness without overhauling the percentage-based rule, while preserving the audience’s decision-making power.

The system is built on three core principles:

- **Audience-Driven Principle:** The elimination results must primarily be determined by audience voting, with the overall logic remaining highly consistent with the percentage-based rule.
- **Competitive Integrity Principle:** Contestants with exceptional technical performances should not be eliminated due to short-term fluctuations in popularity.
- **Broadcast-Friendly Principle:** The rules must be transparent, automatically enforced, and avoid judges’ discretion, allowing the host to easily explain them.

8.2 Structure of the ATCE System: A Three-Tier Progressive Framework

The ATCE system follows a “three-tier progressive framework,” with each level logically connected to form a closed loop. Each tier has a clear objective and operational standards, ensuring clarity and feasibility.

The decision-making process unfolds in three steps: protective mechanisms, consensus judgment, and adaptive integration. If judges and the audience agree on the weakest contestant, they are eliminated immediately; if not, a “two-person showdown” occurs, with judges’ and audience weights adaptively merged based on technical disparity. This design balances the protection of strong contestants, respect for audience preferences, and fairness.

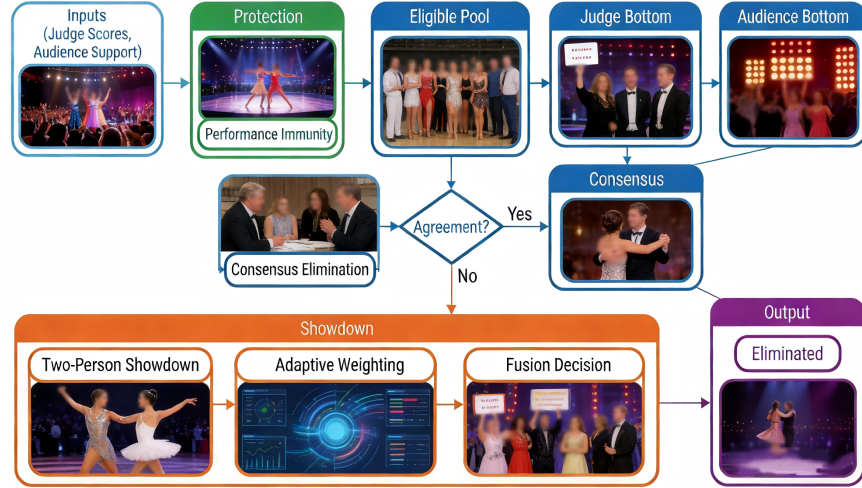


Figure 14: ATCE Decision-Making Process Diagram

8.2.1 First Level: Double Threshold Protection Mechanism (DTPV)

Objective: Mitigate risk for top contestants while preserving the audience’s dominant role and preventing unjust eliminations.

Implementation:

- **Immunity Eligibility:** Contestants’ standardized weekly scores are calculated:

$$Z_{i,t} = \frac{J_{i,t} - \mu_t}{\sigma_t}$$

If $Z_{i,t} \geq \theta_{\text{safe}}$, immunity is granted.

- **Elimination Pool Protection:** If too many contestants are immune, only the top 3-4 contestants with the highest scores receive immunity, ensuring a sufficient elimination pool.
- **Season-Level Protection:** Contestants with a low average technical index over several weeks are ineligible for the final, preventing low-skilled contestants from winning based on popularity.

8.2.2 Second Level: Consensus-Based Elimination Mechanism (TLE)

Objective: Simplify the elimination process when judges’ and audience opinions align.

Implementation:

- **Elimination Pool:** Exclude immune contestants to form the pool:

$$C_t = N_t \setminus I_t$$

- **Identify Consensus Worst Contestant:**

- Judges' worst: $i_J = \arg \max_{i \in C_t} R_{i,t}^J$
- Audience's worst: $i_F = \arg \max_{i \in C_t} R_{i,t}^F$

- **Consensus Elimination:** If $i_J = i_F$, the worst contestant is eliminated.

8.2.3 Third Level: Adaptive Percentage Fusion in Case of Disagreement

Objective: Minimize intervention while protecting strong contestants and respecting audience preferences when there is disagreement.

Implementation:

- **Pending Contestants:** If $i_J \neq i_F$, the two contestants enter a “two-person showdown.”
- **Adaptive Weight Calculation:** Judges' weight is adjusted based on the standardized score difference between the contestants:

$$\Delta_t = \sigma_t(J_{i_J,t} - J_{i_F,t})$$

The weight ranges from $\alpha_{\min} = 0.50$ (minimal disparity) to $\alpha_{\max} = 0.75$ (large disparity), with smooth adjustment using a sigmoid function.

8.3 Empirical Validation: The Necessity and Moderation of ATCE

Using the counterfactual simulation framework constructed in the previous sections, we validate ATCE from three perspectives: feasibility, fairness, and trade-off balance.

8.3.1 Feasibility: High Consistency with Historical Elimination

“Week-by-week match probability” measures how well each rule aligns with actual historical eliminations. This metric assesses the acceptability of the rule change (the closer to 1, the more likely it will be adopted by the producers).

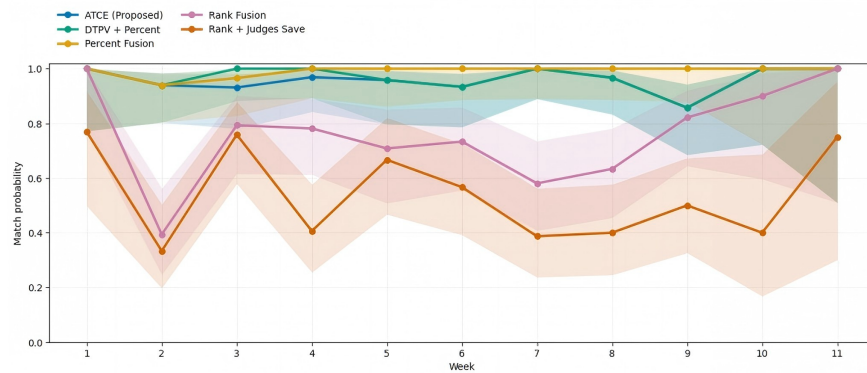


Figure 15: Week-by-Week Match Probability Curve

The curve in the figure shows the probability of matching each week's elimination result with the historical outcome, with a 95% confidence interval shaded. ATCE shows a high consistency with DTPV+Percent in most weeks, deviating only in a few key weeks, indicating that its intervention is minimal and targeted.

8.3.2 Fairness: Significantly Reducing Unjust Eliminations of Top Contestants

We define the “unjust elimination rate” as the probability that a contestant eliminated in a given week is ranked among the top 25% or 33% of all contestants based on judges’ scores. This indicator measures how well the system protects technically skilled contestants.

ATCE and DTPV+Percent significantly reduce unjust eliminations, thanks to immunity, consensus elimination, and adaptive weighting, offering fairness without disrupting audience control.

8.3.3 Comprehensive Trade-Off: Positioned on the Pareto Frontier for Optimal Balance

We use the Pareto frontier to assess the balance between feasibility and fairness. The closer to the bottom-right corner, the better the trade-off between the two.

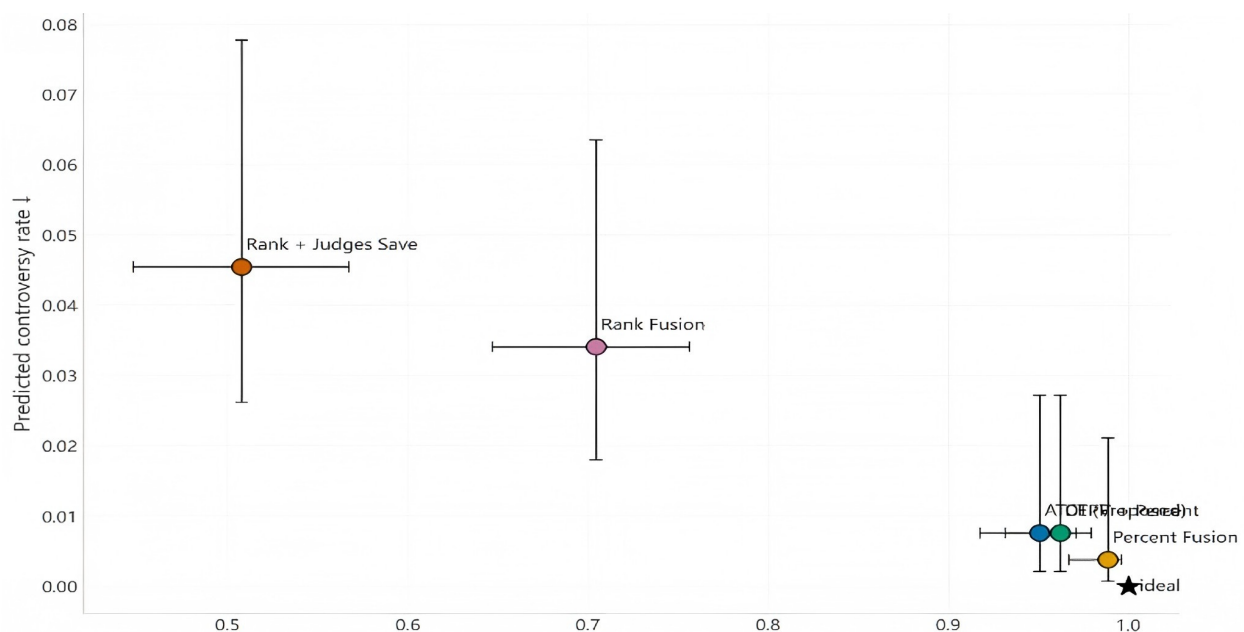


Figure 16: Pareto Frontier Diagram

The horizontal axis represents consistency with historical elimination results (the higher, the more adoptable), and the vertical axis represents predicted controversy (the lower, the more fair). The results show that ATCE achieves a better balance, significantly reducing controversy without sacrificing consistency, positioning it near the Pareto frontier, indicating an optimal balance between feasibility and fairness.

8.4 Producer Perspective: The Adoptability of ATCE

From the practical needs of television program production and broadcasting, ATCE offers the following three core advantages over other rules:

- **Transparency and Reduced Manipulation:** Fully rule-based operation eliminates discretionary manipulation and enhances public trust.

- **Narrative-Friendly, Aligning with Broadcasting Needs:** Concepts such as “consensus elimination” and “two-person showdown” are easily communicated and naturally heighten dramatic tension.
- **Aligning with the Show’s Core Focus:** By preserving audience primacy and applying only restrained technical safeguards, ATCE maintains DWTS’s core identity as a competition shaped by its viewers, balancing entertainment value with competitive legitimacy.

9 Strengths and weaknesses

9.1 Strengths

Accurate Rule Detection:

The model uses a diagonal Gaussian-based structural change detection framework to identify the key point in Season 28 where the voting rule shifted from percentage to ranking, aligning perfectly with historical data and providing a reliable temporal benchmark.

Robust and Interpretable Vote Estimation:

The latent fan vote inference model replicates historical elimination sequences with over 96% Top-1 accuracy. It outputs key metrics like elimination probabilities and certainty, adhering to the principle of “equal weighting” between judges and the audience, enhancing interpretability.

In-Depth Analytical Framework:

The Monte Carlo counterfactual framework isolates the effects of the ranking versus percentage rules, quantifying the amplification of fan influence and evaluating the regulatory role of “judge rescues” with high precision.

Beyond Prediction—Mechanism Discovery:

Through fan-judge synergy analysis and rule mining, we reveal structural patterns like “dual-channel resonance” and “popularity compensation,” enhancing the model’s predictive accuracy and cross-season robustness.

Feasible System Balance:

The **Adaptive Threshold-Consensus Elimination (ATCE)** system maintains transparency, technical protection, and audience dominance, reducing unjust eliminations while aligning with historical logic. Its automation and transparency make it highly applicable in live broadcasts.

9.2 weaknesses

Generalization Relies on Stable Assumptions:

The model assumes stable historical voting patterns. Structural changes in audience behavior or contestant types could reduce its generalizability. Integrating real-time data (e.g., social media) would enhance its adaptability.

Uncertainty in Extreme Case Estimates:

Probability estimates for extreme cases, like Bobby Bones, are based on limited simulations, introducing uncertainty in low-probability, high-impact events.

Feature Set Could Be Expanded:

Current features focus on structured data, excluding factors like emotional appeal, choreographic narrative, and public sentiment, limiting the model's ability to fully explain the complexity of fan support.

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Memo

To: *Dancing with the Stars* Production Team

From: MCM Team 2626177

Subject: Recommendations for Optimizing Vote Synthesis and Competition System

Date: February 3, 2026

Dear Producers,

Based on Seasons 1–34 data, we evaluated the synthesis of judges' scores and audience votes, focusing on fairness, entertainment, and robustness. Below are our findings and recommendations.

I. Key Findings

Vote Estimability and Rule Evolution:

Season 28 marked the shift from **percentage** to **ranking** methods, with a 96.97% Top-1 prediction accuracy in our model.

Differences Between Synthesis Rules:

The **percentage method** amplifies audience influence, while the **ranking method** reduces this effect, aligning better with judges' assessments.

Early Signals and Partner Influence:

A “fan-judge synergy” emerges early, with consistent evaluations ensuring stable progression, while conflicts lead to controversial eliminations. Dance partners' past performance significantly affects scores and audience support.

ATCE Proposal and Validation:

We propose **ATCE**, a three-tier elimination system that balances **audience control** and **technical protection** for contestants. Simulations show ATCE maintains consistency with historical outcomes and reduces unfair eliminations.

II. Recommendations

Phased Rule Design:

Start with **percentage voting** to boost engagement, then switch to **ranking + judge interventions** post-semifinals for fairness.

Implement ATCE for Long-Term Optimization:

ATCE corrects extreme cases without disrupting the existing system. Its **transparent** and **automated** nature makes it ideal for live broadcasts, retaining dramatic moments like “two-person showdowns.”

Enhance Early Signal Interpretation:

Use the **fan-judge synergy index** to show the “technical-popularity” alignment of contestants, improving viewer understanding.

Dynamic Parameter Adjustment:

Adjust ATCE's thresholds and weight parameters each season to match specific dynamics.

III. Conclusion

The program's success lies in balancing audience engagement with professional integrity. We recommend piloting ATCE's mechanisms to gradually optimize the system, enhancing fairness and long-term viewer loyalty.

We are available for further analysis if needed.

Best regards,

Team #2626177

Report on Use of AI Tools

1. ChatGPT (GPT-5.2)

Query: <On the website <https://datausa.io/>, look up the GDP value of Washington when the thirteenth season of "Dancing with the Stars" was broadcast.>

Output:<According to my search on the content provided by the DateUSA website, I found that the GDP of Washington state when the thirteenth season of "Dancing with the Stars" was broadcast was 377,518.80 million dollars.>

2. ChatGPT (GPT-5.2)

Query:<On the website <<https://www.bea.gov/>>, look up the population of Pennsylvania when the twenty-sixth season of "Dancing with the Stars" was broadcast.>

Output:<According to my search on the content provided by the BEA website, I found that the population of Pennsylvania when the twenty-sixth season of "Dancing with the Stars" was broadcast was 12.8M.>

3. DeepSeek (February 2025 Edition)

Query: <Please rewrite the following conclusion in a more objective and neutral academic tone: "The two-segment model is fitted to the candidate change point cp , and its goodness of fit is evaluated by the piecewise log-likelihood. A larger value of this statistic indicates a stronger explanatory power of the dual Gaussian distributions for the observed data, and thus a higher probability that the candidate point corresponds to the true change point.">

Output: <Here, $\hat{\mu}_j$ and σ_j^2 denote the sample mean and variance of the j -th feature, and n represents the number of seasons in the segment. For a given candidate breakpoint, the total log-likelihood equals the sum of the log-likelihoods of the two segments. A higher value indicates stronger evidence that the candidate point corresponds to a true structural change in voting rules.>

4. DeepSeek (February 2025 Edition)

Query: <Please rewrite the following description of "uncertainty modeling" to be more concise and highlight the method logic:

"The estimation of fan latent variables involves uncertainty, and its weekly fluctuation range can be adjusted through the 'identifiability' indicator.">

Output: <We compare the championship distribution driven by fan preferences (P) with that under fully random fan voting (Q):>

5. DeepSeek (February 2025 Edition)

Query: <Rewrite the following suggestion to make it more persuasive and structurally clear:

"In practice, a more robust solution is: to balance audience enthusiasm and competition fairness, the percentage combination method can be used in the early season to maintain heat, and Judges Save can be introduced in the late season as a safety valve.">

Output: <A hybrid design—percentage rules early and Judges Save in later stages—offers a robust balance. Thus, voting synthesis rules reflect deliberate trade-offs among fairness, participation, and stability rather than neutral technical choices.>